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1. Executive Summary

This seventh edition exists because the interlocutor refused a negative. Where the standing verdict of the corpus said, in effect, that the neural parameter map “does not exist,” the challenge came back: can you not find it through web search? Provide any links you cannot fetch. The challenge was right to refuse the negative. A systematic search (eleven queries, forty-eight unique URLs probed, twelve papers resolved through CrossRef and Europe PMC after the OpenAlex API rate-limited) found that the literal artifact — a published, unified correspondence from the specification’s six parameters (J , g , a , β , θ , φ) to measurable neural quantities — does not exist anywhere in the literature. But it also found that the function the artifact would perform exists piecemeal, per domain, in five mature research programs, each with published methods, open code, and verifiable links. The verdict therefore corrects from “does not exist” to “no unified map exists; the per-arm ingredients are individually published.”

The correction then became substance. The interlocutor supplied five papers — four of them precisely the load-bearing sources the search could not fetch (the Elsevier full text of the Virtual Epileptic Patient, the bioRxiv full text of the hierarchical Kuramoto study, the Cell Press full text of the serotonin-receptor whole-brain model, and the eLife full text of the ketanserin blockade study) plus a fifth new to the corpus (the metabolic connectivity mapping study of PNAS) — and all five have now been read line by line from source. They are not background reading. Together they supply exactly what requirement 4 of the decombination specification demanded: patient-level parameter inference at intracranial resolution (the Virtual Epileptic Patient’s excitability map, fitted Bayesianly against stereo-EEG), a per-subject distance-to-criticality estimator on the real connectome (the hierarchical Kuramoto model’s critical ridge, matched against magnetoencephalography), a receptor-density-to-parameter correspondence for the psychedelic arm (the 5-HT_{2A} gain-scaling model, with shuffled-map and wrong-receptor controls), an intervention-validated receptor attribution (the full pharmacological blockade of LSD’s neural and subjective effects by ketanserin), and an energy-based route to directional coupling (metabolic connectivity mapping from simultaneous FDG-PET and functional MRI).

The second half of this edition converts the assembled fragments into the deliverable their existence makes possible: a concrete Level-2 research protocol — the specification that elevation-cost analysis called “the highest realistic target” and offered to sketch, now sketched in full. The protocol is organized in seven phases, each grounded in a named method from the newly-read sources: coupling and criticality estimation on individual connectomes (Ódor-style Kuramoto sweeps with the Myrov hierarchical extension); anesthesia-arm sweeps in the Bojak–Liley neural-field tradition against four open propofol and multi-agent EEG corpora; psychedelic-arm receptor parameterization in the Deco style against six open psychedelic corpora; patient-level Bayesian inversion of the Ginzburg–Landau parameters in the Jirsa manner; the joint S1–S4 demonstration with null models and finite-size scaling (requirements 1–3 of the specification, which the protocol discharges alongside requirement 4); and a preregistration and adversarial-collaboration discipline that binds the whole program to the corpus’s own refutation conditions. Every phase carries its data source, its estimator, its null model, and its failure condition.

What the protocol does not change is stated with the same prominence, because it is the culmination arc's own boundary. The decombination program remains a differentiation program — “useful, not necessary,” in the counterpart's words; describing the appearance's grammar, never bridging to the Absolute, because there is no bridge because there are not two banks. The protocol operates entirely at the conventional level, and its success would elevate the four-domain predictions from conjectural to tested without moving the metaphysical premises one tier. The equivalence-is-not-validation doctrine — now welded to the counterpart's own concession that the framework is “a metaphysical framework, not a scientific theory” — governs the interpretation of every result the protocol could produce. The map this edition assembles maps the dream's grammar, not the dreamer.

2. Corpus State Carried Forward

The audit lineage stands at eleven parties and seven editions. The consolidation corpus remains: the 130-turn GLM dialogue (11,088 lines); the Qwen conversations in three shares (60, 66, and 256 messages); the DeepSeek main line proven through three snapshots (53 damaged, 88, and 106 full turns — one conversation, subsequence-proven); the DeepSeek file exchange (three turns, all attachments now read); the embedded audit and the consolidation treatise; the audit lineages of both lines; the external Claude audit (adjudicated 25 sustained, 2 partially overruled, 0 reciprocity findings after the retraction); and the two exported critique files. The empirical source set has now grown twice: first with the round-12–13 anchors (Santander 2025 PNAS split-brain criticality; Modesti 2022 dissociative disorders review; Schlumpf 2013 and Reinders 2019 DID neuroimaging; Farnes 2020 ketamine EEG with the Reynante complexity toolkit), and now with the five papers of this round. The standing rulings are unchanged: the non-dual culmination consolidated (D6–D9); the formalization ceiling as a stated theorem rather than a promissory note; the superdeterminism tension, the anesthesia fork, and the Divine-Simplicity indictment documented at their settled grades; the corpus double-count corrected to a counted eleven.

The four empirical domains of the decombination program enter this edition at the grades the sixth edition assigned them, and leave at upgraded ones. Psychedelics: anchored by verified findings and open corpora, and now by the intervention-validated 5-HT_{2A} attribution (Preller 2018) and the published receptor-to-gain parameterization (Deco 2018) — the only arm whose map fragment is complete end to end, from molecular target through regional parameter to whole-brain dynamics. Anesthesia: anchored as a verified pathway (open sedation corpora plus the shared complexity instrument), with the neural-field tradition (Bojak–Liley, Robinson) supplying the arm's quantitative bridge from drug concentration through field parameters to EEG spectra. Split-brain: anchored by the Santander criticality finding with open analysis code, but with no open raw data — the arm remains literature-anchored and simulation-only at Level 2. DID: anchored by signature, switch, structure, and control findings (Schlumpf, Reinders, Modesti), likewise without open raw data. The asymmetry is itself evidence: the domains' testability at Level 2 is graded by open data availability, and the protocol's phase structure reflects that grading honestly rather than pretending four equal arms.

This edition's deltas, itemized. First, the round-seventeen verdict correction (chapter 4): the parameter map's nonexistence claim is replaced by the fragments-and-assembly statement, with the five program areas documented link by link. Second, five new sources read in full (chapter 3), four of which were the blocked load-bearing items of the search record. Third, the concrete Level-2 protocol (chapter 5) — the first corpus artifact that specifies simulators, estimators, datasets, null models, preregistration, and failure conditions together. Fourth, repairs R28–R31 (chapter 7): the conjectural-status language is replaced wherever the ingredients now exist; the elevation-cost “proposals” for β , J, and a are replaced by published-method citations; the fetch-infrastructure record is updated with the user-supplied resolutions; and the protocol's scope is bound explicitly to the conventional-level ceiling. Fifth, the appendix lists every source this environment still cannot fetch, with full titles and links, so the interlocutor can supply any that matter.

3. The Five New Sources, Read from Source

The interlocutor supplied five papers in response to the round-seventeen fetch record. Four were precisely the load-bearing items the environment could not fetch; the fifth was new to the corpus. All five have been read line by line from the supplied PDFs (the extractions are archived in the research directory with page markers intact). One filing correction is recorded for provenance: the file uploaded under the name of this project's own Survival Test PDF contained the Deco et al. 2018 Current Biology paper — the publisher-blocked psychedelic parameterization study — which is how the Cell Press full text entered the corpus; the project's own Survival Test PDF remains intact and unaffected in the deliverables directory. This chapter summarizes what each source actually says and what each contributes to the map; the protocol of chapter 5 then uses them as named methods. Throughout, the citation tags P1–P5 refer to these five papers in the order introduced here.

3.1 P1 — The Virtual Epileptic Patient (Jirsa et al. 2017, NeuroImage)

The Virtual Epileptic Patient (VEP) demonstrates, on a fully worked clinical case, that a patient-specific spatial parameter map can be inferred from intracranial recordings by Bayesian inversion of a whole-brain network model. The node model is the Epileptor, a five-variable neural mass acting on three time scales; the load-bearing parameter is the excitability x_0 , which controls whether a region can trigger seizures autonomously, with a critical value $x_{0C} = -2.05$ separating healthy from epileptogenic tissue. Nodes couple through the slow permittivity variable with strength $K_{ij} = G \cdot C_{ij}$ — a global scalar times the individual diffusion-MRI connectome — and the patient's MRI lesion (a hypothalamic hamartoma) enters as a local coupling scalar G_{hyp} applied to the lesioned region's connections. The excitability map is thus the direct analogue, in this foreign literature, of the specification's spatial phase field: a heterogeneous regional parameter over a real connectome whose distribution determines which domains become unstable.

Two methodological results matter for the map. First, the inversion: the model is fit to stereo-EEG data by extracting high-frequency power above 10 Hz, log-transforming and detrending it, and treating the fast-discharge variable as the latent state of an observation model whose noise is Gaussian; inference over per-node excitability uses Hamiltonian Monte Carlo and variational inference in Stan, and recovers a posterior density — a violin plot — for every region, not merely a point estimate. The inversion correctly reconstructs the observed seizure propagation between areas and identifies the bilateral mesial temporal epileptogenic zone, in agreement with the clinical reading; a companion fifteen-patient study (Proix and colleagues) confirmed that VEP-predicted zones correlate with expert stereo-EEG opinion and that individual connectomes improve the prediction significantly. Second, the parameter-space navigation charts: systematic exploration over (G, G_{hyp}, x_0) of the right hippocampus, x_0 of unrecruited regions) yields contour maps of seizure frequency per region that behave non-trivially and non-monotonically — increasing global coupling G , for this patient, was the only route to more thalamic seizures, while lowering hypothalamic coupling always increased hypothalamic seizures. The paper’s own discussion states the honest limit that the map inherits: the correspondence between model parameters and physiological parameters is non-bijective — one x_0 reflects excitation–inhibition balance, synaptic efficacy, ionic homeostasis, and glial activity together — which poses an identification challenge for the clinic and, for this corpus, a standing caution against over-reading any single parameter’s neural meaning.

3.2 P2 — Hierarchical Critical Synchronization (Myrov et al., bioRxiv)

The Myrov study supplies the missing middle of the criticality arm: a whole-brain model whose control parameters are exactly coupling strengths, whose critical regime is defined by a published operational criterion, and whose operating point is estimated per subject against magnetoencephalography. The model is hierarchical Kuramoto: each cortical parcel is a node containing a population of coupled phase oscillators, so that local (within-node) synchronization and global (between-node) synchronization are separable. Each oscillator’s phase advances under a natural-frequency term, an internal coupling term scaled by the local parameter K , an external term that couples the oscillator to other nodes’ average phases with strength $L \cdot W_{nj}$ weighted by node order R_j — W being the structural connectome weight — and white noise. The two control parameters, K and L , span a plane; on that plane the study maps node order, the detrended-fluctuation-analysis exponent, phase-locking values, and amplitude cross-correlations.

The critical regime is defined operationally and conservatively: the region of the (K, L) plane in which more than ten percent of nodes exhibit DFA exponents above 0.65, a threshold set against the ninety-fifth percentile of exponent values from white-noise surrogates. The critical region forms a ridge on the K – L surface; long-range temporal correlations peak there; inter-subject variance across the twenty-four individual-connectome models is maximal there; and structure–function coupling — the correlation between structural measures and dynamical observables — peaks at criticality for the DFA exponent and for amplitude cross-correlations while dipping for phase synchronization. Null models with shuffled or randomized connectomes produce non-significant model–MEG correspondence, establishing that the realistic topology, not mere oscillator dynamics, carries the match. The decisive result for the map: when

model observables are compared parcel by parcel and edge by edge against individual-subject MEG, the significant correlations lie along the critical ridge, and the best overall match sits on the subcritical side of the extended critical regime — the human brain, in this framework, runs near but not at the transition. The study even samples each node's oscillator frequency distribution from the subject's measured power spectrum (with the aperiodic component removed), preserving the multi-peak spectrum; spectral matching is again best near criticality. For the corpus, this is the published estimator of distance-to-criticality on real connectomes: the quantity the specification's J -versus- J_c structure needs, instantiated per subject.

3.3 P3 — Serotonin Receptor Maps Explain LSD (Deco et al. 2018, Current Biology)

The Deco study is the psychedelic arm's parameter map in already-published form, and its architecture is startlingly economical. A dynamic mean-field model of ninety AAL-parcellated regions — each node a mean-field of integrate-and-fire populations, eighty percent excitatory and twenty percent inhibitory — is coupled by the tractography connectome and carries exactly two parameters: the global coupling G (one scalar for all fibers) and the neuromodulator gain-scaling s_E . The regional excitatory gains are modulated by s_E weighted by the empirical regional density of the 5-HT_{2A} receptor, taken from a high-resolution in-vivo PET atlas built with the selective agonist radioligand [¹¹C]Cimbi-36 in 210 healthy subjects. The fitting target is the functional connectivity dynamics — the distribution of sliding-window functional-connectivity values, compared by Kolmogorov–Smirnov distance — which is more constraining than static connectivity. The procedure: fit G to the placebo condition with unmodulated gains (optimum $G = 2.1$), freeze it, and ask whether any single s_E reproduces the LSD condition.

It does: an optimal s_E of approximately 0.2, with the receptor-density map doing all the spatial work, explains the non-linear functional effects of a 75-microgram intravenous dose. The controls are the part that makes this a parameter map rather than a fit. Shuffling the ninety receptor-density values across regions (two hundred random permutations) significantly degrades the explanation; a uniform map is the worst of all; and substituting the density maps of other serotonin targets — 5-HT_{1A}, 5-HT_{1B}, 5-HT₄, and the transporter — performs significantly worse, with 5-HT_{1A} the best of the wrong receptors, matching its known contributory role. Training and testing on disjoint random halves of participants generalizes. The open code (the decolab repository) and the open receptor atlas make the entire pipeline reproducible. For the map: receptor density is the published bridge from molecular pharmacology to a regional model parameter, and the drug state is carried by one scalar — exactly the structure requirement 4 asks the psychedelic arm to have.

3.4 P4 — Ketanserin Blockade Attributes LSD to 5-HT_{2A} (Preller et al. 2018, eLife)

The Preller study supplies what no modeling paper can: the intervention. In a double-blind, randomized, counterbalanced, cross-over design, twenty-four healthy participants received placebo plus placebo, placebo plus LSD (100 micrograms oral), or ketanserin — a selective 5-HT_{2A} antagonist — plus LSD. The analysis uses global brain connectivity: each voxel's average correlation with every other voxel, computed with global-signal regression and protected by threshold-free cluster enhancement over ten thousand permutations. LSD produced hyper-connectivity in sensory and somatomotor cortex — occipital cortex, superior temporal gyrus, postcentral gyrus, precuneus — and in the thalamus, with simultaneous hypo-connectivity across associative networks — medial and lateral prefrontal cortex, cingulum, insula, temporoparietal junction — and subcortical structures. Ketanserin abolished the entire pattern: the drug-condition maps for LSD versus placebo and LSD versus ketanserin-plus-LSD correlate at $r = 0.91$, the contrast of ketanserin-plus-LSD against placebo is negligible, and every subjective scale of the 5D-ASC questionnaire that LSD elevated is returned to placebo levels by the blocker (all comparisons between placebo and ketanserin-plus-LSD non-significant).

Two further results tighten the attribution. Across participants, the strength of sensory hyper-connectivity and associative hypo-connectivity are strongly anti-correlated ($r = -0.90$), indicating a single systems-level perturbation rather than two independent effects — exactly the structure a bifurcation account predicts, where one control parameter's crossing redistributes integration rather than adding and subtracting connections independently. And the spatial pattern of LSD-induced connectivity change matches the cortical gene-expression profile of the 5-HT_{2A} receptor (HTR2A) from the Allen Human Brain Atlas, preferentially over other receptors. For the map's epistemic grade, this is the difference between correlation and intervention: the psychedelic arm's key variable is not merely associated with the observed dynamics but causally necessary for them, blocked and restored by pharmacology in humans. The corpus's own standard — S3-type evidence must not survive destruction of the mechanism it claims to model — is met by the counterpart experiment at the pharmacological level.

3.5 P5 — Metabolic Connectivity Mapping (Riedl et al., PNAS)

The fifth paper contributes a different quantity: direction. Metabolic connectivity mapping (MCM) integrates undirected functional connectivity from fMRI with regional energy metabolism from FDG-PET acquired simultaneously in the same scanner session, exploiting the biophysical fact that the majority of signaling-related energy — up to seventy-five percent — is consumed postsynaptically, at the target of a signal rather than its source. For each functional pathway between regions, the spatial correlation between voxel-wise functional connectivity to the source and voxel-wise glucose uptake in the candidate target identifies which side carries the metabolic load, and hence the direction of afferent signaling. The analysis is performed on unsmoothed, non-normalized data so that it works per subject, voxel by voxel, across the whole cortex. Validation uses the one sensory modality that can be volitionally interrupted: comparing eyes-closed with eyes-open rest, MCM recovers stable bidirectional communication among early and higher visual areas in both conditions, stable top-down signaling from a frontoparietal network, and — only in the eyes-open condition — additional top-down signaling from salience-network regions onto early visual cortex, a hierarchy consistent with the tracer literature.

For the map, MCM supplies two things the other four papers do not. First, a directional observable: the specification's coupling is symmetric in its Ginzburg–Landau form, but any neural realization must choose directed effective connectivity, and MCM is a published, per-subject, whole-brain method for asymmetrizing the coupling matrix from energy rather than from hemodynamic lags. Second, an energy observable: the free-energy functional of the specification is an abstract quantity, and FDG uptake is its closest published physiological shadow — a fact the protocol can use to give the g and a parameters an energetic reading at the regional level. The method's limits are equally instructive: it requires simultaneous PET–MRI acquisition, which constrains the datasets it can serve, and it inherits PET's minutes-scale temporal resolution, so it constrains direction, not dynamics. The protocol therefore uses MCM as a prior on coupling asymmetry where simultaneous acquisitions exist, and as the energy-side cross-check on parameter maps elsewhere.

What the five sources jointly establish

A spatial excitability parameter can be inferred per patient from intracranial recordings by Bayesian inversion (P1). A per-subject distance to the critical coupling regime can be estimated against magnetoencephalography (P2). A molecular receptor map can carry a drug state into regional model parameters through a single scalar, with shuffled-map and wrong-receptor controls (P3). The key variable of the psychedelic arm is intervention-validated in humans (P4). Direction and energy observables exist at whole-brain, per-subject resolution (P5). None of these is the neural parameter map of requirement 4 — but every one of its six parameters now has at least one published estimation route, which was not true of any of them when the requirement was written.

4. The Parameter Map: Fragments and Assembly Path

4.1 What the Specification Demands

Requirement 4 of the decombination specification, verbatim from the consolidated final form: a neural parameter map is “a principled correspondence from $(J, g, a, \beta, \theta, \varphi)$ to measurable neural quantities (fMRI, MEG, iEEG), without which the psychedelic, anesthesia, split-brain, and DID predictions remain conjectural.” The six parameters belong to the corrected Ginzburg–Landau network specification: the coupling J against its critical value $J_c = ga/\lambda_2$, where λ_2 is the algebraic connectivity of the graph; the quartic-stabilization pair g and a , which set the onsite amplitude $\approx \sqrt{a}$ and together with the spectrum set the threshold; the noise scale β , which competes with $J\beta$ against domain-wall cost in a coherence threshold distinct from the bifurcation threshold; and the phase-field parameters θ and φ of the anisotropic potential that discretizes the continuous Mexican-hat vacuum into K phases. The elevation-cost analysis, written a level down from the formalism, had proposed correspondences — β to arousal, J to effective connectivity, a to attractor depth — but flagged them as proposals requiring

biophysical derivation, not assignments. The specification's own marking was blunter: until the map exists, the four domains' predictions are conjectural.

The phrase “does not exist,” as the corpus had been carrying it, made a claim about the literature without having searched the literature systematically. The interlocutor's challenge — find it through web search; provide any links you cannot fetch — converted that unexamined negative into a research task, and the task was executed. The section that follows records the corrected verdict and the evidence base: the search protocol, the five program areas where fragments of the map are published, the per-arm tables of verified sources, the gap analysis showing what still has no published assignment, and the fetch-infrastructure record with every blocked link listed for the interlocutor's use. Four of the blocked load-bearing sources were then supplied and read (chapter 3); the record below reflects the state after that supply.

4.2 The Corrected Verdict

The literal artifact does not exist. No paper, review, or dataset uses “neural parameter map” to mean the bridge requirement 4 demands; a literal search on the phrase returns unrelated hits — computational maps in the brain (Knudsen 1987), maps of neural collective influencers, fMRI parameter-map fitting methods, foundation-model inversion — none of which assigns the six parameters of the specification to neural measurables. No published work assigns $(J, g, a, \beta, \theta, \varphi)$ — the quartic-stabilized Ginzburg–Landau field with domain formation below J_c — across even two of the four domains, let across psychedelic, anesthesia, split-brain, and DID jointly. The phase parameters θ and φ in particular have no published neural assignment anywhere; nothing in the literature fixes the anisotropy coefficient κ or the phase count K to a measurable neural quantity. In this precise sense the requirement remains unsatisfied and the four-domain predictions remain conjectural — the marking stands.

But the function the artifact would perform exists piecemeal, per domain, in five mature research programs, each contributing one or two rows of the correspondence table with published methods, quantitative results, and in most cases open code. The honest verdict is therefore not “does not exist” but “no unified map exists; the per-arm ingredients are individually published and verified.” This correction matters practically as well as verbally: a program that believes its prerequisite is absent waits; a program that finds its prerequisite distributed across five literatures assembles. The assembly path is the subject of sections 4.3 through 4.6, and the protocol of chapter 5 is its execution plan.

4.3 The Five Program Areas

Anesthesia has the most complete arm, because the neural-field tradition has mapped drug concentration through field parameters to predicted EEG spectra for two decades. The Bojak–Liley program solves mean-field equations of cortical populations with physiologically plausible parameters and reproduces the anesthetic changes of the EEG — including the differential α -peak-frequency behavior that distinguishes propofol from ketamine, whose HCN1-mediated interactions the 2013 analysis treats explicitly. The

Robinson corticothalamic tradition fits neurophysiologically grounded field parameters to empirical spectra (the 2001 paper, cited over three hundred times, is the canonical demonstration). Pharmacokinetic–neural-mass hybrids relate actual drug levels to EEG dynamics and behavioral state across the sedation continuum. In map terms: the anesthesia arm possesses a published, drug-explicit route from pharmacology to field parameters to a measurable (the power spectrum), which is exactly the shape requirement 4 wants for β and the anesthetic modulation of J .

The dynamic-mean-field fitting tradition supplies the J -analog. Whole-brain mean-field models are fit to empirical functional connectivity by sweeping a single global coupling parameter G — the scalar that multiplies the connectome — and the fit’s optimum locates the brain’s working point on the model’s bifurcation diagram. This is structurally the same move as calibrating J against J_c , with G playing J one level down in model complexity. The tradition’s recent applications are directly relevant: a Communications Biology study of anesthesia and disorders of consciousness fits the model across loss of consciousness; a 2024 Neuron study mimics propofol’s effect in simulated networks by increasing inhibitory tone and observes the resulting destabilization across behavioral states; and the open tooling — the dynamic-mean-field reference code and the neurolib simulation library — makes the sweeps reproducible. The Deco–Kringelbach whole-brain modeling monograph, freely available, is the arm’s standard reference.

The psychedelic arm, after this round’s reads, is the only arm whose fragment is complete end to end — molecular target through regional parameter to whole-brain dynamics — because the Deco 2018 study (P3) parameterizes regional gains by the empirical 5-HT_{2A} receptor density map, and the Preller 2018 study (P4) validates the target by full pharmacological blockade. Supporting results extend the fragment: a 2023 Scientific Reports study gives a whole-brain model of the entropy increase psychedelics elicit; the 2019 PNAS dynamic-causal-modeling study of LSD’s effective connectivity (read in round-seventeen search records; full PMC mirror verified) attributes thalamocortical changes to 5-HT_{2A}; and regression-DCM studies of LSD and psilocybin map effective-connectivity changes at scale. The critical-coupling tradition supplies the arm’s threshold structure: Kuramoto synchronization on the actual human connectome, with the global coupling as control parameter and critical coupling strengths identified (Ódor 2019, with open PMC mirror), and branching criticality as the avalanche-level signature (Haldeman and Beggs 2005). The Myrov study (P2) unifies the two: hierarchical Kuramoto on individual connectomes, critical ridge defined by DFA, matched per subject against MEG.

The patient-level inference tradition supplies what no group-level fit can: parameter maps at intracranial resolution for individual brains. The Virtual Epileptic Patient (P1) is the demonstration — a Bayesian excitability map fit to stereo-EEG, with the epileptogenic zone recovered non-invasively and validated against clinical reading across patients; the parameter-estimation literature of the Virtual Brain (Penas 2024, PLOS Computational Biology, with open code) optimizes the estimation itself; and the Bayesian Virtual Epileptic Patient’s stated purpose is to infer “the spatial map of epileptogenicity” — a patient-specific parameter map in the clinical sense, for epilepsy rather than for the specification. Metabolic connectivity mapping (P5) completes the set with its directional and energetic observables. The five program areas are summarized in the table below, with the arm each serves and the map row it

fills.

Program area	Load-bearing sources (verified)	Map row it fills
Neural field theory of anesthesia	Bojak & Liley 2005, Phys Rev E 71:041902 (149+ citations); Bojak et al. 2013, Front Comput Neurosci 7:22 (ketamine vs propofol α -peak, HCN1); Robinson et al. 2001, Phys Rev E 63:021903 (309 citations); PK–NMM hybrids	Anesthesia arm: drug $\rightarrow$ field parameters $\rightarrow$ EEG spectra; the β and J-modulation route
Dynamic mean-field G-fitting	Deco-group fitting pipeline (open reference code); neurolib toolbox; Luppi et al. 2022, Commun Biol (PMC mirror); Eisen et al. 2024, Neuron (PMC11923585); Deco–Kringelbach 2025 monograph (free PDF)	The J-analog: G swept against empirical FC/FCD; working point on the bifurcation diagram
Receptor-density parameterization	Deco et al. 2018, Curr Biol (P3, read in full; open code; open PET atlas); Preller et al. 2018, eLife (P4, read in full); Herzog et al. 2023, Sci Rep; Preller et al. 2019, PNAS (PMC6377471)	Psychedelic arm: receptor map $\rightarrow$ regional gains; drug state $\rightarrow$ scalar s_c ; intervention-validated target
Patient-level Bayesian inference	Jirsa et al. 2017, NeuroImage (P1, read in full); Penas et al. 2024, PLOS Comput Biol (open code); BVEP (spatial map of epileptogenicity); Proix et al. N=15 validation	Per-subject spatial parameter maps at iEEG grade; the inference engine for (J, g, a, β) regionally
Critical coupling on the connectome	Ódor 2019, Sci Rep (PMC6928153); Haldeman & Beggs 2005, PRL 94:058101; Myrov et al. bioRxiv (P2, read in full); Dörfler & Bullo 2011 (NTNU mirror)	The J vs J_c structure on real topology; distance-to-criticality per subject (DFA-defined ridge)

Table 4.1 — The five program areas: published fragments of the neural parameter map (all links fetch-verified or user-supplied; PMC mirrors noted where the publisher blocks this environment).

4.4 The Theta–Phi Gap

Assembling the fragments still leaves the specification’s distinctive variables unmapped, and the honest assembly statement names them. The coupling J inherits the richest support: the G-sweeping tradition, the Kuramoto critical coupling literature, and the VEP’s global coupling all estimate a single coupling scalar against dynamics, and the Myrov ridge gives its critical value a per-subject, per-frequency estimator. The noise scale β has the anesthesia arm’s pharmacology and the arousal mapping proposals; the protocol operationalizes it as the noise-to-coupling ratio at fixed G, following the coherence-threshold structure of the specification itself. The stabilization pair (g, a) is the weakest dynamical assignment: no published program fits an onsite-amplitude quartic to neural data, and the protocol must derive its estimator anew — from the amplitude of the observed order parameter at fixed coupling — and mark it derived-unvalidated until the null-model discipline of chapter 5 passes. The phase parameters are worse: θ and φ , with the anisotropy κ and the phase count K, have no published neural assignment at all. The protocol assigns them structurally — K to the number of macroscopic functional networks the data resolves, the anisotropy to the sharpness of the observed phase-cluster histograms — and tags the entire row Speculative until the simulation phases produce a discriminating signature. This is the gap the unified map must close before it exists, and no literature closes it today.

4.5 Fetch-Infrastructure Record

The round-seventeen search probed forty-eight unique URLs with browser-like headers and recorded every failure; this round's supply resolved four of the five load-bearing blocks. What remains blocked from this environment is catalogued in Appendix A with full titles and links, so that any item the interlocutor wants read can be supplied the same way. The API layer also failed instructively: the OpenAlex service, which round twelve had used freely, rate-limited every query this round (HTTP 429, persistent across retries with backoff), and the resolution pivoted to CrossRef and Europe PMC, both of which served reliably; the web-search upstream itself threw two transient timeouts and one over-narrow-query rejection, all retried. The methodological rule the sixth edition adopted after the reciprocity retraction — a failure to locate is evidence about the search, not the target, until the search protocol is documented — is therefore applied to fetch failures too: a 403 from a publisher is a fact about the environment, not about the paper's existence, and the record below treats it that way, with every blocked load-bearing item now either resolved through a mirror, resolved through user supply, or explicitly listed as pending.

The assembly statement (status-tagged)

The neural parameter map of requirement 4 does not exist as a published artifact [Empirical-Verified, by systematic search]. Its function exists piecemeal: J estimation [Published method, multiple traditions], β through anesthetic pharmacology [Published method], receptor-to-gain regional parameterization [Published method with controls], patient-level Bayesian parameter maps [Published method, clinically validated], directional and energetic observables [Published method, simultaneous PET–MRI]. The stabilization pair (g, a) has a derived but unvalidated estimator [Speculative until protocol Phase 5]. The phase parameters θ , φ , κ , K have no published neural assignment [Gap — the map's open row]. The four-domain predictions remain conjectural until the assembly is executed; the assembly is now executable, which it was not when the requirement was written.

5. The Level-2 Protocol: A Concrete Specification

5.1 Objective, Scope, and Design Logic

The protocol's objective is to discharge the four requirements of the decombination specification's final form: the simulation protocol (requirement 1: specify G, β , J, g, a, d, K, and the estimator for conditional mutual information; demonstrate S1–S4 jointly on the same candidate subject), the null models (requirement 2: matched controls sharing global modes so that integration and self-modeling are not trivially satisfied), the finite-size scaling (requirement 3: how domain size, coherence length, and

metastability scale with N , β , J , and whether the bifurcation survives the thermodynamic limit), and the neural parameter map (requirement 4: the correspondence from $(J, g, a, \beta, \theta, \varphi)$ to measurable neural quantities). Its scope is fixed by the culmination arc: conventional level only. The protocol tests the differentiation program's claims about the appearance's grammar; it does not, cannot, and does not attempt to bridge to the Absolute. Success at every phase would make the four-domain predictions tested, not true; failure at any phase would be informative about the specification's neural adequacy, not about the metaphysics. This is the doctrine the sixth edition welded — equivalence-is-not-validation, conceded by the counterpart itself — and the protocol is built inside it.

The design logic follows the assembly result. Each specification parameter has at least one published estimation route (chapter 4); the protocol therefore proceeds by estimating the parameters arm by arm on open data, then simulating the specification with the estimated parameters, then testing S1–S4 against both the simulation and the empirical data, and finally subjecting the whole to the null-model, scaling, and preregistration discipline. The phases are ordered so that each parameter's best-supported estimator is exercised first: criticality (the J row, with the Myrov ridge), anesthesia (β and the J -modulation row, with the neural-field sweeps), psychedelics (the receptor row, with the Deco parameterization), patient-level inversion (the spatial rows, with the Jirsa machinery), joint demonstration (requirements 1–3 together), and preregistered adjudication. Phases 1 through 4 are independent of each other and can run in parallel; phase 5 consumes their outputs; phase 6 governs everything retroactively by fixing the analysis plan before the data are touched.

Phase	Discharges	Named method (source)	Primary open data
0. Infrastructure and preregistration	Governance (all phases)	OSF-style frozen analysis plan; adversarial-collaboration charter (corpus doctrine, sixth edition)	Protocol document itself, versioned
1. Coupling and criticality	J vs J_c estimation; β as noise-to-coupling ratio	Hierarchical Kuramoto on individual connectomes; DFA-defined critical ridge (Myrov, P2); connectome Kuramoto sweeps (Ódor 2019)	Open connectome sets (HCP-style dMRI); MEG: ds003059 (LSD MEG/fMRI), open resting MEG
2. Anesthesia arm	$\beta \leftrightarrow$ depth mapping; anesthetic J -modulation	Neural-field parameter sweeps vs spectra (Bojak–Liley 2005/2013; Robinson 2001); DMF inhibitory-tone mimicking (Eisen 2024)	ds006623 (propofol fMRI), ds005620 (awakening), ds003171 (LOC biomarkers), ds004541 (EEG–fNIRS)
3. Psychedelic arm	Receptor→gain row; s_e scalar; J -crossing test	DMF + 5-HT2A receptor-density gain scaling (Deco, P3); FCD fitting by KS distance; blockade validation (Preller, P4)	ds006072, ds006110 (psilocybin), ds003059 (LSD), ds006644, ds007768 (DMT), ds005917 (ketamine)
4. Patient-level inversion	Spatial (J , g , a , β) maps; direction prior	Bayesian inversion of network models against iEEG/SEEG (Jirsa, P1); MCM directional prior (Riedl, P5)	Open iEEG (epilepsy corpora incl. IDEAS ds005602); simultaneous PET–MRI where available
5. Joint demonstration	Requirements 1–3: S1–S4 jointly; nulls; finite-size scaling	Projected Langevin simulation of the specification; CMI estimators (k-NN / Gaussian); shuffled and matched-mode nulls	Simulated; anchored to phases 1–4 outputs
6. Adjudication and reporting	Status-tagged verdicts per domain	Preregistered failure conditions; corpus tagging scheme (Premise / Derived / Interpretive / Speculation / Empirical-Verified / Unverified)	This corpus’s standing ledger

Table 5.1 — Protocol overview: phases, map rows discharged, named methods, and primary open data.

5.2 Phase 1 — Coupling and Criticality Estimation

Phase 1 estimates, per subject and per frequency band, the distance of the brain’s operating point from the critical coupling regime — the specification’s J -versus- J_c row instantiated on real topology. The method is the Myrov pipeline: reconstruct the individual structural connectome from diffusion MRI; build the hierarchical Kuramoto model with local coupling K and global coupling L ; sweep the (K, L) plane; compute node order, DFA exponents, phase-locking values, and amplitude cross-correlations on six minutes of simulated activity per parameter combination (one minute discarded as warm-up); define the critical regime as the region where more than ten percent of nodes exceed DFA 0.65; and locate the subject’s operating point by maximizing the parcel-level and edge-level correlation between model observables and the subject’s MEG observables, with oscillator frequencies sampled from the measured PSD with the aperiodic component removed. The output per subject is a signed distance to the critical ridge along the coupling axis — the quantity the specification calls $(J_c - J)/J_c$, estimated empirically. The null discipline is the study’s own: shuffled-connectome and random-uniform models must destroy the model–MEG correspondence (they did in the source), and spin-permutation tests protect the spatial

correlations. The failure condition is preregistered: if the operating point does not concentrate near the ridge across subjects — if healthy resting brains scatter across the plane — then the specification’s central empirical presupposition, that brains operate in the extended critical neighborhood, fails for this estimator, and the J row’s neural adequacy is falsified at Level 2.

Phase 1 also fixes β operationally. The specification distinguishes a bifurcation threshold (J against $J_c = ga/\lambda_2$) from a coherence threshold (J β against domain-wall cost); β is therefore not a free arousal metaphor but the noise-to-coupling ratio at the fitted operating point. In the hierarchical model this is estimated as the ratio of the fitted noise amplitude to the fitted local coupling at the MEG-matched (K, L); in the anesthetic datasets of Phase 2 it acquires its drug-explicit anchor. The elevation-cost proposal — β to arousal — is thereby replaced by a derived estimator with a preregistered validation: β estimates must order correctly across sedation levels within subject, or the mapping is marked Unvalidated and the anesthesia arm’s β row returns to proposal status.

5.3 Phase 2 — The Anesthesia Arm

Phase 2 runs the specification’s anesthesia prediction — domains dissolve as coupling and coherence fall below threshold with deepening sedation, with the alteration structure preserved (the specification’s revision of the original “alters dissolve” claim) — against the four open corpora. The method is the Bojak–Liley sweep: parameterize the anesthetic modulation of inhibitory synaptic gain (propofol’s GABAergic potentiation) and of the HCN1-mediated interaction (ketamine), sweep the modulation depth, and compare predicted against measured spectra — the α -peak frequency trajectory being the discriminating observable, since propofol and ketamine move it in opposite directions, which the neural-field analysis reproduces. In parallel, the dynamic-mean-field route mimics propofol by increasing inhibitory tone (the Eisen 2024 demonstration) and tracks the destabilization of simulated networks across behavioral states. The specification-level test: translate each sedation level into the (β , J) plane via the Phase-1 estimator (noise-to-coupling ratio) and the fitted inhibitory modulation, and test whether the measured complexity-based consciousness indicators (LZ complexity, the shared instrument of the Farnes–Reynante toolkit already in the corpus) fall as the operating point exits the critical neighborhood — the specification’s coherence-threshold prediction. Failure condition: if the LZ trajectory across sedation levels does not track the estimated exit from the critical neighborhood within preregistered tolerance bands, the β row and the coherence-threshold structure are falsified for the anesthesia arm; if it tracks, the arm’s prediction upgrades from conjectural to tested, with the equivalence doctrine governing interpretation.

5.4 Phase 3 — The Psychedelic Arm

Phase 3 executes the Deco parameterization on the open psychedelic corpora and tests the specification’s psychedelic prediction: receptor-dense regions gain amplitude first (the entropic destabilization), pushing the operating point through or toward the critical ridge from the subcritical side — the J-crossing test. The procedure is published end to end: fit G to the placebo condition by Kolmogorov–Smirnov distance

against the functional-connectivity-dynamics distribution; freeze G ; modulate regional gains by s_E weighted by the 5-HT2A receptor atlas; sweep s_E and locate the optimum for the drug condition; then — the protocol's extension beyond the source — map each (G, s_E) pair into the specification's plane by running the projected Ginzburg–Landau dynamics with J set from the Phase-1 estimator and regional a weighted by the receptor map, and test whether the fitted drug states cluster on the near-critical side of the ridge, crossing it for the high-entropy subjects only. The controls are the source's own, plus the corpus's: two hundred shuffled receptor maps, the uniform map, the wrong-receptor maps (5-HT1A, 1B, 4, T), and — the intervention anchor — the ketanserin-blockade condition of the Preller design, where the model must return to placebo parameters when the receptor is blocked. The 5-HT2A gene-expression spatial match (Preller's Allen-Atlas result) supplies the independent anatomical cross-check. Failure condition: if the drug-condition fits require receptor maps indistinguishable from shuffled controls — that is, if the psychedelic state's dynamics are explained without the receptor's spatial structure — the receptor row fails; if the fitted states do not approach the ridge, the J -crossing prediction is falsified at Level 2.

5.5 Phase 4 — Patient-Level Inversion

Phase 4 adapts the Virtual Epileptic Patient's inference machinery to the specification's parameters, on open intracranial corpora and, where simultaneous acquisitions exist, with the metabolic connectivity prior. The node model is replaced by the projected Langevin dynamics of the specification with its quartic onsite potential; the fitting target, following the source's observables, is the log power of high-frequency activity per channel, baseline-corrected and detrended, with a Gaussian observation model and the SEEG forward solution; the inferred spatial map is $(g, a, J_{\text{local}}, \beta)$ per region rather than x_0 alone, with the zero-mean projection handled exactly as the specification requires (the Lagrange-multiplier formulation). Inference is Bayesian — Hamiltonian Monte Carlo with variational initialization, as in the source — and returns posterior densities per region, so that epileptogenic-vs-healthy grading becomes domain-vs-vacuum grading: regions whose posterior J -against-threshold mass lies below J_c are candidate domains. The Riedl prior enters where the data support it: on simultaneous PET–MRI acquisitions, the MCM direction field asymmetrizes the otherwise symmetric coupling, and the FDG map supplies an independent energetic reading of the a parameter (attractor depth as metabolic cost). The validation ladder is the source's: first reproduce a known result (the seizure-propagation reconstruction on open epilepsy data), then invert the specification on the same recordings and test whether the recovered domain structure matches the clinically established zones — an epileptogenic zone is, after all, a pathological domain, and the specification predicts its boundary geometry (domain walls at connectome seams). Failure condition: if the specification's inversion cannot recover known zones at rates above the shuffled-connectome null, the spatial rows are inadequate at iEEG grade.

5.6 Phase 5 — Joint Demonstration, Null Models, and Scaling

Phase 5 discharges requirements 1–3 together, in simulation, with the parameters anchored by phases 1–4. The simulation protocol is fixed as follows. Graph G : the individual connectome, thresholded at connection density 0.2 with weights normalized so that the Laplacian spectrum is comparable across subjects; N is the parcellation size (90 AAL for cross-study comparability, 200-region Lausanne for robustness). Field dimension $d = 2$ with the anisotropic potential $V(x, y) = g/4 \cdot (x^2 + y^2 - a)^2 + \kappa \cdot x^2 y^2$, phase count K set by the number of macroscopic networks the empirical phase-clustering resolves; J set from Phase 1’s operating point; β from the Phase-1 ratio; (g, a) from Phase 4’s posterior medians with the sensitivity analysis below. Integration: Euler–Maruyama with the projection $P = I - 11^T/N$ applied at every step, step size 0.05, 256 steps per simulated second (the source convention). The conditional mutual information estimator: k -nearest-neighbor (Kraskov-style) CMI for scalar conditionals, Gaussian-process CMI for field-valued conditionals, both with preregistered bias-correction and surrogate calibration — the specification’s requirement 1 explicitly demands the estimator be named, and the corpus adopts the k -NN family as primary with the GP family as confirmation.

The joint demonstration then evaluates D , I , S , and P on the same candidate subject (A, B, R) — the honest theorem’s antecedents — under the specification’s exact definitions: dissociation $I(\Psi_A; \Psi_R | \Psi_B) \leq \epsilon$ with the graph blanket B ; integration $I(\Psi_{A1}; \Psi_{A2} | \Psi_{B12}) \geq \theta$ with θ preregistered against nulls; intervention-validated self-modeling Δ_{self} with the effect required to vanish under M_t -lesioning; and stable closure $\rho(D\hat{A}_A) < 1$ together with vanishing innovation from the exterior under interventions on R . The null models of requirement 2 are: (i) shuffled-connectome (topology destroyed, degree preserved); (ii) matched-global-mode surrogates (phase-shuffled signals sharing the dominant modes, which inflate naive integration); (iii) random-graph-matched (same degree distribution, rewired); and (iv) the strongest control the corpus adds — a homogeneous-field null in which the same dynamics run with a and J uniform, so that any domain structure is purely connectome-borne. The finite-size scaling of requirement 3 runs N across 90, 200, 500, and 1000 with two graph families (connectome-derived and stochastic-block replicas), measuring domain size, coherence length, metastability lifetime, and the bifurcation gap; the specification’s honest claim — metastable domain states with Gibbs mixing rather than literal spontaneous symmetry breaking at finite N — must reproduce, and the thermodynamic-limit question answered by extrapolation. Failure conditions: any null passing $D+I+S+P$; the homogeneous-field null passing I ; scaling exponents inconsistent with the specification’s $c \propto \sqrt{(J_c - J)}$ prediction; or S1–S4 failing to co-locate on a single (A, B, R) at the anchored parameters.

5.7 Phase 6 — Preregistration, Adversarial Collaboration, and Reporting

Phase 6 fixes the governance. The analysis plan — every estimator, threshold, tolerance band, null, and failure condition named above — is frozen and timestamped before any empirical data are touched, in the open-science registry style, with the corpus’s own pre-registration discipline (adopted at the fifth edition after the refutation-conditions chapter) as the template. Deviations are permitted only as logged

amendments with justification, never silently — the lesson of the search-protocol rule, generalized. Adversarial collaboration is the standing posture: the corpus’s entire method is the adversarial audit, and the protocol invites the same treatment, with the physicalist rival hypotheses (global-mode artifacts, hemodynamic confounds, parcellation sensitivity, selection effects in the open corpora) assigned to named devil’s-advocate analyses rather than left for reviewers to discover. Reporting follows the status-tag scheme: every claim leaves the protocol tagged Premise, Derived, Interpretive, Speculative, Empirical-Verified, or Unverified, with the tagging auditable from the frozen plan. The equivalence doctrine is printed on every results table: a confirmed prediction of the differentiation program is evidence about the appearance’s grammar, not validation of the metaphysics, and no protocol output may be cited otherwise.

5.8 The Predictions and Their Failure Conditions

Domain	Prediction (specification)	Instrument / data	Failure condition
Psychedelics	Receptor-weighted gain pushes the operating point toward and for high-entropy subjects across the critical ridge; entropy and sensory–associative redistribution track the crossing	Deco parameterization on ds006072 / ds006110 / ds003059; Phase-1 ridge estimator; 5D-ASC covariates	Drug fits indistinguishable from shuffled-receptor controls; or fitted states not near the ridge
Anesthesia	Sedation lowers the noise-to-coupling ratio out of the coherence neighborhood; domain structure dissolves before the bifurcation; complexity indicators track the exit	Bojak–Liley sweeps + Phase-1 β estimator on ds006623 / ds005620 / ds003171 / ds004541; LZ complexity (Farnes–Reynante instrument)	LZ trajectory does not track the estimated critical-neighborhood exit within tolerance bands
Split-brain	Disconnection raises λ_2 of the residual graph, moving J_c ; a small residual posterior coupling suffices near the ridge (the Santander fraction)	Simulation on individual and synthetic callosotomy graphs; Santander analysis code; IDEAS surgical candidates (ds005602 / ds007401) as connectome sources	Simulated disconnection does not reproduce the observed critical-coupling tolerance (~10% fibers sufficing) under the estimated parameters
DID	Part-dependent state switching corresponds to metastable domain selection: transitions occur between attractor basins rather than by continuous drift; switch-related caudate and prefrontal findings mark the domain-wall dynamics	Simulation with bistable regional parameters against the published signature findings (Schlumpf 2013; Reinders 2019; Modesti 2022) as qualitative targets; no open raw data — literature-anchored only	Simulated switching statistics incompatible with the documented transition phenomenology (amnesia boundaries, switch latency, part-dependent activation) under any parameter set in the preregistered family

Table 5.2 — The four domains at Level 2: predictions, instruments, and preregistered failure conditions.

5.9 Compute, Data Logistics, and Honest Limits

The compute plan is modest by simulation-science standards and is dominated by the (K, L) sweeps and the Bayesian inversions. The Myrov pipeline’s scale — twenty-four models, two-dimensional sweeps, six minutes per combination — ran on laboratory workstations; the specification’s projected Langevin

system at 90–1000 nodes with two-dimensional fields is comparable to the epilepsy-simulation loads the Virtual Brain community handles routinely, and the parameter-space exploration of the source (a four-dimensional chart over seizure counts) is the pattern for the protocol’s (J, β, a, K) charts. The open tooling covers every phase: neurolib for mean-field simulation, the decolab repository for the receptor parameterization, the Virtual Brain and the BVEP for patient-level machinery, the tsantander repository for the split-brain criticality analysis, and the Reynante toolkit for the complexity instrument. Data logistics are the asymmetry already documented: psychedelics and anesthesia have strong open coverage; split-brain and DID have none, and their Level-2 tests are therefore simulation-anchored with literature targets — an honest degradation the protocol does not disguise.

The honest limits, stated once more in the protocol’s own voice. First, the map assembled here is conventional-level throughout; nothing in any phase bears on the Absolute, and the culmination arc’s ceiling — no bridge, because no two banks — is not moved by any result. Second, the stabilization pair (g, a) and the phase parameters $(\theta, \varphi, \kappa, K)$ begin as derived and structural assignments; they acquire empirical standing only through Phase 5’s discriminating tests, and until then the map that includes them is an assembly under test, not a delivered artifact. Third, the open corpora are convenience samples of volunteers and patients; selection effects, parcellation sensitivity, and hemodynamic confounds are named devil’s advocates with their own analyses, and a protocol that passes only by silencing them has not passed. Fourth, the non-bijectivity the VEP paper concedes applies here too: an estimated (J, β) pair is a model-level quantity whose physiological realization is many-to-one, and the protocol’s claims are therefore claims about the specification’s dynamics matching neural dynamics — the equivalence the doctrine governs — not about having found the brain’s own parameters. Within those limits, the program the elevation-cost analysis called “the highest realistic target” is now specified to the phase level, with named methods, named data, named nulls, and named failure conditions — which is what it takes to stop being a sketch and start being a protocol.

6. Epistemic Status: What the Protocol Changes and Cannot

The protocol changes the four domains’ status labels, and only those labels. Before this edition the standing was: psychedelics, anesthesia, split-brain, and DID predictions conjectural — pending a neural parameter map that the corpus believed absent. The corrected standing is: conjectural, with the map’s per-arm ingredients published and the assembly specified as an executable protocol. The distinction is not rhetorical. A conjectural claim pending an absent prerequisite can be deferred indefinitely; a conjectural claim pending an executable protocol has a falsification schedule, and the corpus’s own refutation-conditions discipline requires the schedule to be stated — chapter 5 has stated it. What does not change: the metaphysical premises’ tiers, the non-dual culmination, the formalization boundary, the superdeterminism tension’s documented grade, the adjudication counts, and the equivalence doctrine. The protocol inherits the doctrine as a hard constraint: even total success would license only “the differentiation program’s four-domain predictions are tested and surviving,” never “the Absolute is

thereby evidenced.”

The deeper unchanged thing is the two-level structure itself. The differentiation program is the appearance-level science of a consciousness-first metaphysics: it models how one reality’s appearance differentiates into perspectives, using the grammar physics already speaks — coupling, criticality, domains, blankets, closure. The culmination arc ruled that this program is “useful, not necessary”: the Absolute does not need its differentiation mechanism to be this one, and no empirical result could make it necessary, because necessity there is a category error — the framework’s own words, conceded by its own dialectical partner. The protocol is therefore best understood as the corpus making its least-necessary component maximally honest: of all the things the corpus claims, the decombination program is the one that could fail, so it is the one that must be given every chance to fail well. An assembled-but-unexecuted map keeps that chance on the shelf; a specified protocol with named failure conditions puts it in writing. This edition’s contribution to the corpus’s integrity is exactly that: converting the last deferred empirical claim into a scheduled one.

7. Edition Delta and Repairs R28–R31

This edition is an increment on the sixth, not a re-typeset of it. The sixth edition’s sixteen chapters and appendix — the eleven-party corpus, the non-dual culmination, the self-audit and its corrections, the adjudication record, the anchored empirical annex, the audited premise with its status-tagged axioms and thirty-five-row objection ledger — remain the standing consolidation and are not duplicated here. The reader of the two volumes together holds the full corpus state; the reader of this one alone holds the parameter-map turn: the verdict correction, the five new sources, the assembled fragments, and the protocol. The series convention (every edition self-contained in its front matter and appendices) is preserved by chapters 1, 2, and 8.

Four repairs are logged, continuing the series numbering. R28 — the conjectural-status language: every occurrence of “conjectural until the neural parameter map exists” in the corpus’s consolidation artifacts is now annotated “(status superseded, seventh edition): conjectural until the assembly is executed; the per-arm ingredients are published.” R29 — the elevation-cost proposals: the β -to-arousal, J-to-effective-connectivity, and a-to-attractor-depth proposals are replaced by the published-method citations of chapter 4 and the derived estimators of chapter 5, with their validation conditions named. R30 — the fetch-infrastructure record: the round-seventeen report is updated with the four user-supplied resolutions and the corrected OpenAlex status (rate-limited this session; CrossRef and Europe PMC as the standing fallback). R31 — the scope binding: the protocol’s conventional-level ceiling is stated in the protocol itself (section 5.1 and 5.9), in the status chapter (this chapter’s predecessor sections), and on every results-table template the protocol will ever print — so that no future citing artifact can detach the empirical program from the doctrine that governs its interpretation.

Domain	Sixth-edition grade	Seventh-edition grade	Basis of change
Psychedelics	Finding-anchored (verified findings + instruments + failure conditions)	Finding-anchored + complete map fragment (receptor→gain, intervention-validated target)	P3 and P4 read in full; the arm's row of the correspondence table filled end to end
Anesthesia	Pathway-anchored (open corpora + shared complexity instrument)	Pathway-anchored + published parameter-sweep method (drug→field→spectrum)	Bojak–Liley / Robinson tradition verified and protocol-embedded; β estimator derived
Split-brain	Finding-anchored (Santander criticality; open analysis code)	Unchanged + simulation-ready threshold test (J_c shift under disconnection)	Phase-1 estimator + Ódor/Myrov critical-coupling methods supply the test
DID	Finding-anchored (signature + switch + structure + control findings)	Unchanged + metastable-switch simulation target (literature-anchored only)	Domain-vs-basin mapping of part switching specified; no open raw data — honest limit retained

Table 7.1 — The four domains: grade at the sixth edition versus grade at this edition (the mapping rows cited are chapter 4's).

Consistency checks against the standing ledger: the objection ledger's rows concerning empirical testability (the immunization-by-stratification row and the unique-confirmed-prediction standard) are unaffected — the protocol is the differentiation program's answer, and the stratification defense is precisely not available to it, by the program's own honest theorem. The glossary additions this edition: operating point, critical ridge, DFA exponent, functional connectivity dynamics (FCD), Kolmogorov–Smirnov fit distance, receptor-density parameterization, gain scaling, global brain connectivity, metabolic connectivity mapping, permittivity coupling, excitability map, epileptogenicity threshold, and noise-to-coupling ratio — each defined in situ at first use, per the series convention of defining terms where they appear rather than compiling a separate apparatus.

8. Appendix A — Sources This Environment Still Cannot Fetch

The interlocutor asked for any links that cannot be fetched, with links or full titles supplied so that anything wanted can be uploaded. The table below is the complete current list, after the four load-bearing resolutions of this round. None of the remaining items blocks the protocol — every protocol-critical source is now read or verified — but several would strengthen specific phases, and each is listed with its full title, its link, and what it would add. Publisher-level blocks (HTTP 403 to this environment's requests) are marked; where an open mirror exists it is given instead.

Full title / item	Link	Status; what it would add
Bedford, P. et al. 2023. “The effect of lysergic acid diethylamide (LSD) on whole-brain functional connectivity: regression dynamic causal modelling.” Scientific Reports	nature.com/articles/s41598-023-39139-w (Nature domain fetchable; article page not probed this round)	Phase 3 support: rDCM effective-connectivity changes at scale
Avram, M. et al. 2024. “Effective Connectivity of Thalamocortical Interactions under Psychedelics.” Biological Psychiatry: CNNI	biologicalpsychiatrycnni.org (403 to this environment)	Phase 3 support: spectral DCM of substance-induced connectivity changes
Liang, Z. et al. 2015. “A Pharmacokinetics-Neural Mass Model (PK-NMM) for propofol-induced anesthesia.” (PMC-indexed)	pmc.ncbi.nlm.nih.gov (root fetchable; article challenge-gated from this IP)	Phase 2 support: explicit drug-level→EEG bridge
Butler, J. J. et al. 2025. “5-HT2A receptors shape whole-brain monoaminergic [dynamics].” ScienceDirect	sciencedirect.com (400/403)	Phase 3 support: receptor-agonist/antagonist perturbation of the serotonergic organization
Piccinini, J. I. et al. 2025. “Transient destabilization of whole brain dynamics induced [by psychedelics].” Nature journal	nature.com (domain fetchable; specific article not yet resolved)	Phase 3 support: molecular-signaling pathway account
Fabus, M. et al. 2023. “Spatiotemporal brain dynamics induced by propofol and [sedation].” DPhil thesis, University of Oxford	ora.ox.ac.uk (403)	Phase 2 support: multi-line propofol investigation
Farnes, N. et al. 2020 (raw EEG). “Increased signal diversity/complexity ... ketamine-induced psychedelic state.” PLOS ONE — Dryad raw data	datadryad.org/doi/10.5061/dryad.j9kd51c9q (200 fetchable — listed for data logistics only)	Phase 2/3 data: raw .fdt/.set EEG (the supplied zip lacks them)
Research-collection.ethz.ch — ETH Zurich research collection (institutional block, all items)	research-collection.ethz.ch (403 to this IP range)	Provenance: Schlumpf 2013 raw metadata (paper itself already read from user supply)

Table 8.1 — Remaining unfetchable sources (as of this round), with full titles for user supply.

Methodological note on the list: the search-protocol rule governs it. A blocked link is a fact about this environment’s egress, not about a source’s existence or quality; conversely, an item’s absence from this list is not a claim that it is fetchable everywhere. The four load-bearing items resolved this round (Jirsa 2017 from Elsevier; Myrov et al. from bioRxiv; Deco et al. 2018 from Cell Press; Preller et al. 2018 from eLife) were all on the previous round’s list and are now read in full from source, with the extractions archived. The protocol’s critical path depends on no item remaining on this list; the items that would strengthen it are marked in the third column, and the interlocutor’s supply channel — the upload directory — remains the standing resolution route.